

SMART URBAN MOBILITY & TRAFFIC INTELLIGENCE DASHBOARD

Milestone 1 - Data Integration and Data Modeling

Project	Smart Urban Mobility & Traffic Intelligence Dashboard
Milestone	Milestone 1
Tool	Microsoft Power BI Desktop
Dataset	300-record mobility dataset
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Batch	3
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Milestone 1 establishes the analytical foundation of the project through structured data preparation, dimensional modeling, DAX-based measures, interactive filtering and dashboard design.

1. Objective

The objective of Milestone 1 is to build a reliable Power BI foundation for Smart Urban Mobility and Traffic Intelligence. The work covers dataset preparation, data modeling, relationship creation, DAX measure development, interactive filtering and a professional analytical dashboard. The final dashboard is designed to answer practical mobility questions such as traffic volume by hour, route congestion, vehicle composition, weather impact and route-level performance.

2. Dataset Overview

The project uses a 300-record mobility dataset. The central mobility table contains record-level transportation observations and links each observation to date, location, route, vehicle and weather dimensions through business keys. The model is designed to support both high-level KPIs and detailed route analysis.

2.1 Key Fields

Field	Role / Description
Mobility_ID	Unique identifier for each mobility record.
DateKey	Key used to relate mobility observations to the Date dimension.
Timestamp	Date and time of the recorded observation.
LocationKey	Key linking the observation to location attributes.
Route_ID	Route identifier used for route-level analysis.

WeatherKey	Key linking the observation to weather conditions.
VehicleKey	Key linking the observation to vehicle attributes.
Hour	Numeric hour used for hourly traffic trend analysis.
Peak_Period	Traffic period category used for peak-period filtering.
Vehicle_Count	Observed vehicle volume.
Passenger_Count	Observed passenger count.
Avg_Speed_kmph	Recorded average speed in km/h.
Delay_Minutes	Recorded delay in minutes.
Avg_Wait_Minutes	Recorded average waiting time.
Travel_Time_Minutes	Recorded travel time.
Congestion_Index	Congestion indicator used for route/weather comparison.
Utilization_Rate	Vehicle/resource utilization indicator.
Accident_Risk_Score	Accident-risk indicator.
CO2_Emission_Index	Environmental impact indicator.
Data_Quality_Score	Data quality indicator used for validation.

3. Data Preparation & Validation

- The data was checked and prepared before visualization so that analytical results would be consistent and relationships would work correctly.
- **Data-type validation:** identifier fields were retained as text where appropriate; DateKey was treated as a numeric key; Timestamp was treated as Date/Time; counts and measurements were treated as numeric fields.
- **Key validation:** Mobility_ID was treated as the unique record identifier, while DateKey, LocationKey, Route_ID, WeatherKey and VehicleKey were validated as relationship keys.
- **Consistency checks:** categorical values such as vehicle type, peak period and weather were standardized sufficiently for slicers and grouped visuals.
- **Relationship integrity:** dimension keys were checked so that dimensions could sit on the one side of one-to-many relationships.

3.1 Important Data Types

Column	Data Type	Reason
Mobility_ID	Text	Identifier; not used for arithmetic.
DateKey	Whole Number	Stable numeric relationship key.
Timestamp	Date/Time	Supports time-aware analysis.
LocationKey	Text	Dimension identifier.
Route_ID	Text	Route identifier/category.
WeatherKey	Text	Weather dimension identifier.
VehicleKey	Text	Vehicle dimension identifier.
Hour	Whole Number	Supports chronological hourly analysis.
Vehicle_Count	Whole Number	Count aggregation.

Passenger_Count	Whole Number	Count aggregation.
Avg_Speed_kmph	Decimal	Continuous speed measure.
Delay_Minutes	Decimal	Continuous delay measure.
Congestion_Index	Decimal	Continuous congestion measure.

4. Data Modeling & Relationships

The Power BI model follows a Star Schema. FactMobilityTable is the central table containing measurable mobility events, while the dimension tables provide descriptive context. This structure keeps the model clear and makes filtering and aggregation predictable.

From Table	To Table	Key	Cardinality	Cross-filter
Dim_Date	FactMobilityTable	DateKey	1:*	Single
Dim_Location	FactMobilityTable	LocationKey	1:*	Single
Dim_Route	FactMobilityTable	Route_ID	1:*	Single
Dim_Vehicle	FactMobilityTable	VehicleKey	1:*	Single
Dim_Weather	FactMobilityTable	WeatherKey	1:*	Single

Primary key: Mobility_ID uniquely identifies a mobility record.

Foreign keys: DateKey, LocationKey, Route_ID, WeatherKey and VehicleKey connect fact records to their respective dimensions.

The dimensions are placed on the one side because their keys are unique, while multiple fact rows can reference the same dimension record. Single-direction filtering is used to reduce ambiguity and keep filter propagation predictable.

5. Dimension Tables

Dimension	Important Fields	Purpose
Dim_Date	Date, DateKey, Day_Name, Day_Short, Month, Month_Name, Quarter, Year, Is_Holiday, Is_Weekend, Is_Working_Day	Time-based analysis, date slicer and calendar context.
Dim_Location	City, Latitude, Longitude, Location, LocationKey, State, Zone, Population, Population_Density_per_km2, Road_Class, Near_Hospital, Near_Market, Near_School	Geographic and location-level filtering and context.
Dim_Route	Route_ID, Route_Type, Distance_km, Lane_Count, Speed_Limit_kmph, Capacity_Vehicles_per_Hour, Origin_LocationKey, Destination_LocationKey, Public_Transport_Corridor, Toll_Route	Route characteristics and route performance analysis.
Dim_Vehicle	VehicleKey, Vehicle_Type, Typical_Capacity, Ownership_Type, Fuel_Consumption_L_per_100km, Emission_Factor	Vehicle mix, capacity and environmental context.
Dim_Weather	WeatherKey, Weather, Temperature_C, Humidity_pct, Rainfall_mm, Wind_Speed_kmph, Traffic_Impact_Level	Weather context and weather-to-congestion comparison.

6. DAX Measures

Measures were created instead of relying only on raw columns so that results recalculate dynamically according to the current filter context.

Measure	DAX Formula	Purpose
Total Vehicles	SUM(FactMobilityTable[Vehicle_Count])	Total traffic volume.
Total Passengers	SUM(FactMobilityTable[Passenger_Count])	Total passengers carried.
Average Speed	AVERAGE(FactMobilityTable[Avg_Speed_kmph])	Average observed speed.
Average Delay	AVERAGE(FactMobilityTable[Delay_Minutes])	Average delay.
Average Congestion	AVERAGE(FactMobilityTable[Congestion_Index])	Average congestion level.
Average Travel Time	AVERAGE(FactMobilityTable[Travel_Time_Minutes])	Average travel duration.
Average Utilization	AVERAGE(FactMobilityTable[Utilization_Rate])	Average utilization.
Average Wait Time	AVERAGE(FactMobilityTable[Avg_Wait_Minutes])	Average waiting time.
Average Accident Risk	AVERAGE(FactMobilityTable[Accident_Risk_Score])	Average accident-risk indicator.
High Congestion %	COUNTROWS(FILTER(FactMobilityTable, FactMobilityTable[Congestion_Index] >= 60)) / COUNTROWS(FactMobilityTable)	Share of high-congestion observations.
Peak Hour Vehicles	MAXX(VALUES(FactMobilityTable[Hour]), [Total Vehicles])	Highest hourly vehicle volume.
Passengers per Vehicle	DIVIDE([Total Passengers], [Total Vehicles], 0)	Passenger-to-vehicle ratio.
Distinct Routes	DISTINCTCOUNT(FactMobilityTable[Route_ID])	Number of unique routes represented.

Route Efficiency	DIVIDE([Average Speed], [Average Travel Time], 0)	Derived route efficiency indicator.
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Note: The measure set visible in the Power BI model includes additional analytical measures beyond the four headline KPIs, allowing the project to be extended in later milestones.

7. Dashboard Design & Visual Selection

The dashboard uses a clean executive layout with four KPI cards at the top, interactive slicers on the left, analytical visuals in the center, and a route-level table at the bottom. The selection of each visual is based on the analytical question being answered.

Requirement / Visual	Fields Used	Why This Visual?	Status
4 KPI Cards	Total Vehicles; Total Passengers; Average Speed; Average Delay	Provides an immediate executive summary of mobility volume, demand, speed and delay.	Completed
Traffic Volume by Hour - Line Chart	X: Hour; Y: Total Vehicles	Hour is ordered/time-based; a line makes peaks and fluctuations easy to follow. A pie chart would not represent a time trend well.	Completed
Average Congestion by Route - Column Chart	X: Dim_Route[Route_ID]; Y: Average Congestion	Routes are categories; columns provide direct comparison. A line could wrongly imply an ordered route trend.	Completed
Vehicle Distribution by Type - Donut	Legend: Dim_Vehicle[Vehicle_Type]; Values: Total Vehicles	Vehicle types form parts of the total vehicle activity. Donut also provides center space for the total. A pie chart would also work, but the donut is cleaner.	Completed
Weather Impact on Congestion - Column Chart	X: Dim_Weather[Weather]; Y: Average Congestion	Weather conditions are categories and the goal is value comparison. A donut would imply part-to-whole, which is not appropriate here.	Completed
Route Performance Overview - Table	Route_ID; Total Vehicles; Total Passengers; Average Speed; Average Delay; Average Congestion	Charts show patterns; the table provides exact route-level values for detailed comparison.	Completed
Slicers	City; Vehicle Type; Peak Period; Date	Enables interactive segmentation of the dashboard without creating separate reports.	Completed
Header / Background	Navy header; light blue-gray canvas	Creates consistent hierarchy, readability and professional visual identity.	Completed

8. Interactive Features

Slicer	Field	What It Enables
City	Dim_Location[City]	Compare a specific city with the overall mobility picture.
Vehicle Type	Dim_Vehicle[Vehicle_Type]	Focus analysis on a selected vehicle category.
Peak Period	FactMobilityTable[Peak_Period]	Analyze mobility conditions by traffic period.
Date	Dim_Date[Date]	Restrict the dashboard to a selected date range.

When a slicer selection is made, the relationship model propagates the filter to the fact table and the DAX measures recalculate under the new filter context. This is the main mechanism behind dashboard interactivity.

9. Baseline Dashboard KPIs

KPI	Baseline Value	Interpretation
Total Vehicles	226K	Overall vehicle volume in the full selected date range.
Total Passengers	Approximately 1M	Overall passenger demand.
Average Speed	29.59 km/h	Average observed mobility speed.
Average Delay	24.09 minutes	Average recorded delay.

The baseline view is intended to be presented with City = All, Vehicle Type = All, Peak Period = All, and Date = 01-01-2026 to 30-01-2026.

10. Key Analytical Insights

- Traffic volume changes substantially by hour, allowing periods of higher and lower activity to be identified.
- Congestion is not uniform across routes, so route-level comparison can reveal corridors requiring closer attention.
- Vehicle activity is distributed across multiple vehicle types, providing a clear view of the composition of mobility demand.
- Average congestion varies across weather categories, supporting further investigation of external-condition effects on traffic.
- The route table complements visual summaries by exposing exact values for vehicles, passengers, speed, delay and congestion.

11. Dashboard Walkthrough

The dashboard should be read from top to bottom. The KPI cards establish the overall mobility situation. The slicers allow the user to define the analytical segment. The hourly line chart describes when traffic occurs, the route congestion chart identifies where pressure is concentrated, the vehicle donut describes traffic composition, the weather chart compares congestion under different conditions, and the route table provides exact supporting values.

12. Milestone 1 Completion Checklist

Requirement	Completion
Dataset prepared and validated	Done
Dimension and fact tables created	Done
Primary / foreign keys identified	Done
Star schema implemented	Done
One-to-many relationships configured	Done
DAX measures created	Done
4 KPI cards created	Done
Pie / Donut chart created	Done
Line chart created	Done
Comparison charts created	Done
Detailed table created	Done
Interactive slicers configured	Done
Dashboard styling and layout completed	Done

13. Conclusion

Milestone 1 successfully establishes the core analytical foundation of the Smart Urban Mobility & Traffic Intelligence project. The solution combines a structured star schema, validated data types and keys, reusable DAX measures, interactive slicers and a focused dashboard. The result is a dashboard that can move from executive-level mobility KPIs to detailed route-level analysis in a single interactive view.

Future milestones can build on this foundation through advanced route analytics, geospatial analysis, forecasting, anomaly detection and predictive congestion intelligence.

14. Final Dashboard Screenshot

